



Software Architecture for Essential & Accidental Uncertainty

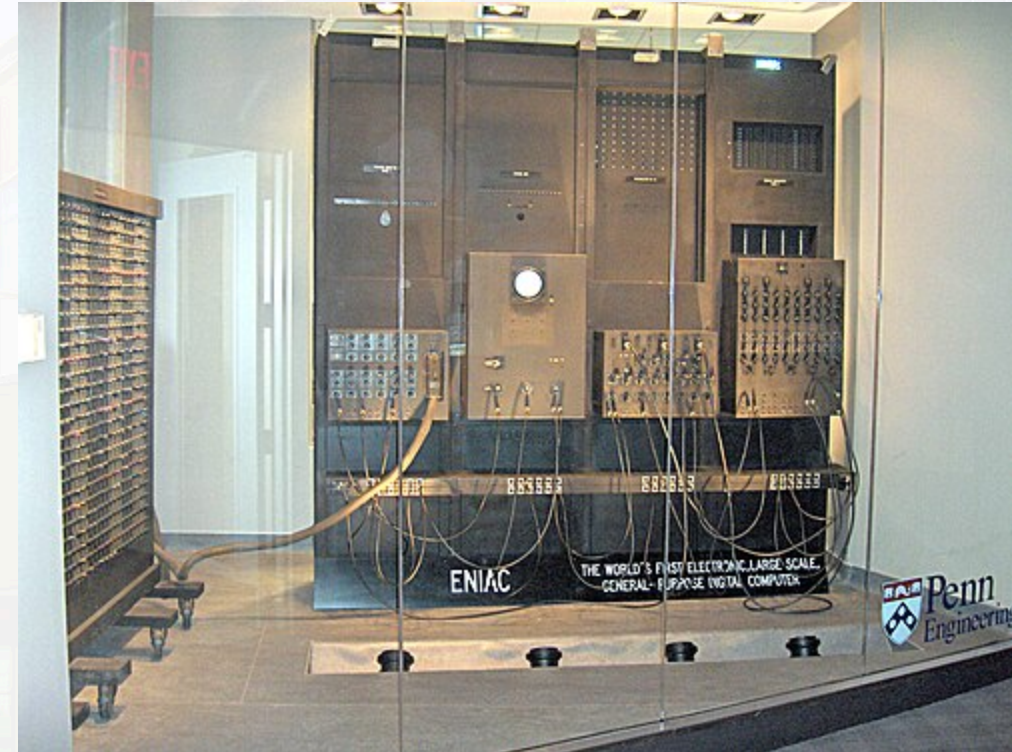
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Introduction: A Bit of History

ENIAC — Electronic Numerical Integrator and Computer

First general-purpose computer built to solve *a large class of numerical problems*. 1945-1955

- First computer to run at electronic speed
- Killed by lightning strike in 1955
- Early problems were **precisely specified**: inputs, algorithms, outputs, ...
- Results were deterministically computed



Progress

- Original computing tackled precisely-defined **mathematical problems**
- Over time, capabilities expanded to **business applications, data communication, control systems, and AI**
 - inputs, outputs, and processes became more varied, less precise.
 - an artifact of modeling non-mathematical problems via mathematical abstractions
- We must now deal with **uncertainty** — and understand how to **approach** it in architecture and design.

Current State of the Art

- (2010) **David Garlan** called for uncertainty to be a first-class concern in software design and development.
- Current research and practice focus on **fault tolerance** and related tactics.
- **This misses a large part of uncertainty** and motivates a deeper investigation.

Essence vs. Accident (Brooks, 1986)

Accidental complexity: created by engineering choices, mitigated by better tools and abstractions.

Essential complexity: inherent in the problem — cannot be removed.

Likewise, there is **essential** and **accidental** uncertainty.



Essential and Accidental Uncertainty

- **Accidental uncertainty:** can be separated from the problem; *a transformation exists* to remove it.
- **Essential uncertainty:** must be addressed as part of the problem; *no transformation* can remove it.

Uncertainty and Data

Computing is the **processing of data**. Consider uncertainty across:

- **Input** (can be essential or accidental)
- **Transformation** from input to output (can be essential or accidental)
- **Output** (often essential)

Input: Uncertainty Examples

- **Unreliable data transmission** — Accidental; datacomm protocols implement transformations to remove uncertainty
- **Amount of data** — Often accidental; handle in subsets
- **User input errors** — Accidental; parse for correctness and re-prompt
- **Multiple concurrent access** — Accidental if handled discretely; **Essential** if not (e.g., venue ticket website, requests for the *same* seat)

UWU Uncertainty in the Transformation (Algorithms)

The **algorithm or processing logic** that transforms input to output can *introduce* uncertainty

- **Accidental** → can be fixed
 - Hardware glitches
 - compiler differences
 - implementation bugs
- **Essential**
 - No deterministic algorithm exists to compute correct output
 - Example: Image recognition
- **Essential (intractable)**
 - Algorithm is computationally infeasible
 - approximations introduce uncertainty in accuracy/precision
 - Examples: Game of Go, optimization problems

Critical insight: Essential transformation uncertainty often reflects **uncertain output** — requires probabilistic or AI-based approaches

Output Uncertainty

- *Accidental* output uncertainty is generally less architecturally interesting.
- **Essential output uncertainty** spans a wide spectrum — requires a corresponding spectrum of architectural approaches.

UvU Essential Output Uncertainties

Output characteristics	Example
There is a known right answer	Mathematical computation
The right answer(s) can be known	Sales website (How well did a product sell?)
Right answer exists but is infeasible to compute directly	Game of Go (astronomical state space)
Unknown right answer; we can guess	Image recognition
Effect is unknown, but can be measured	Optimization problems
Effect is unknown; system is chaotic	Weather forecasting
No specific answer, but can observe "right" direction	Advertising; many simulation problems
No single right answer; the problem is vague	Automated prose writing
The problem itself is unknown	Grounded theory

Thought Question

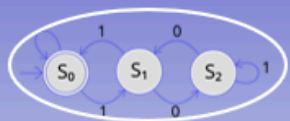
How would you characterize **automatic software coding**?

Trick: It depends on the nature of the software.

If you cannot precisely specify what the generated software should do, **how can you verify it?**

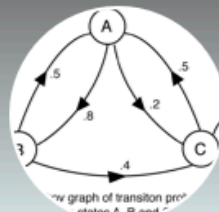
Generative AI makes good answers at once more vital and more complex

Spectrum of Uncertainty



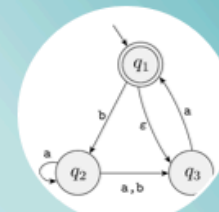
Deterministic

- Exactly one action possible for an input from a given program state
- Always the same action for input, state pairs
- **No uncertainty**



Probabilistic

- More than one action possible for an input from a given program state
- Each transition has a known probability
- Probabilities out of each possible state add to 1
- **Some uncertainty**



Nondeterministic

- More than one action possible for an input from a given program state
- We don't know the transition probabilities
- And/or cannot easily determine possible probabilities
- **High uncertainty**

Less Uncertainty ←

→ More Uncertainty

Architectural Approaches

Different levels and characteristics of uncertainty suggest different architectural styles, patterns, and tactics

- **Virtually no uncertainty** → Batch processing
- **Accidental uncertainty** → Pipes & Filters; Layers; Broker
- **Greater essential uncertainty** → AI-based approaches
 - Blackboard
 - Neural networks
 - Machine learning / deep learning

Spectrum of Architectural Approaches



An Architecture Process for Uncertainty

1. **Identify** uncertainties associated with the system.
2. For each, determine if it is **essential** or **accidental**.
3. **Study the source** of uncertainty (external is often accidental).
4. Determine if a **transformation** can remove the uncertainty.
5. If so, apply **fault-tolerance** and related techniques.

Analyze Uncertainties — Input

- Is it due to **imprecise/incomplete specification**?
- Is it due to **variability in data** (transmission, user errors, ...)?
- What can and cannot be known about input **at runtime**?

Analyze Uncertainties — Output

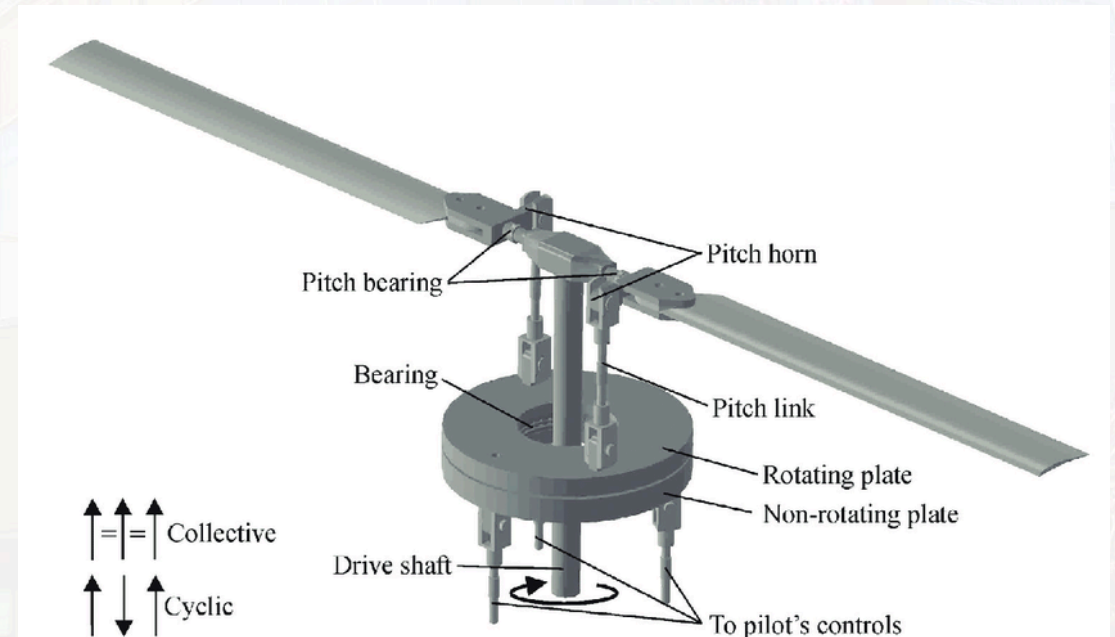
- What is the **objective** of the system? How do we know it meets the objective?
- Is there a **known correct output**?
- Is the correct output **unknowable** (but estimable)?
- Can output be **deterministically** derived from inputs?
- Are there **multiple correct outputs**? If so, is one **preferable**?

Analyze Uncertainties — Transformation

- Is the relationship between input and output **deterministic**?
- If deterministic, is it **tractable**?

Case Study: Helicopter Rotor Balancing

- If the main rotor is not perfectly balanced, excess **vibration** occurs.
- Vibration causes **wear, loss of power**, and can compromise **safety**.
- The rotor is a complex system with multiple **adjustment points**.



Uncertainty Analysis — Input (Rotor)

- Input is **known and precisely measured**: vibration measurements; blade positions; adjustment points.
- **User entry errors** → external; apply standard Fault Tolerance techniques (to correct accidental uncertainty)
- Each helicopter has **unique** wear patterns → *essential*; must be considered.

Uncertainty Analysis — Output (Rotor)

- Objective measurable: **change in vibration** (IPS — inches per second).
- Desired output is a **range** (lower is better).
- Acceptable standard: **0.5 IPS**.
- Single solution? There may be **different sequences** of adjustments achieving the goal.

Uncertainty Analysis — Processing (Rotor)

- Is the relationship between input and output **deterministic**? Initially **unknown**.
- Combinations of adjustments appeared **non-deterministic** / **chaotic**.
- *Initial* Approach: use a **neural network** to derive suggested solutions.

Further Analysis (Rotor)

- **Unique wear profile** for each helicopter.
- Initial NN trained on **many helicopters** → a **generic** approximation.
- **Adjustment/response history** of testing reveals unique profiles.
- **Intermediate Result:** There exists a **deterministic, repeatable** sequence of adjustments with **feedback loops**.
 - details are proprietary

Relevant Results

- System reliably balances main rotors to **0.05 IPS** — *10× better* than the standard.
- Actively marketed to **US Army** and other organizations.
- Demonstrates the importance of understanding **essential vs. accidental** uncertainty.
 - to choose the right architectural approach

Case Study: Online Bookstore

- **Initially:** Inputs/outputs well-specified; numerous **accidental uncertainties** (e.g., unreliable networks).
- Straightforward architecture.

Bookstore Evolution — Related Books

- Add “**related books**” feature.
- Which books are **related**? An **essential uncertainty** — output cannot be known beforehand, no single right answer.
- Approaches become **probabilistic**.

Bookstore Evolution — Promotions

- Use **browsing history** to promote products.
- Even **more uncertainty**; the problem is less well-defined.
- Consider **machine learning**; map to the uncertainty **spectrum**.
- recommendation engines are a popular, common solution.

References

- Harrison, N.B., Rudolph, G., and Aldous, P. *Software Architecture for Essential and Accidental Uncertainty*. HICSS, 2026.
- Harrison, N.B., Rudolph, G., and Aldous, P. *Is Artificial Intelligence the Answer to Everything?* Intermountain Conference on Engineering, Technology and Computing (IETC), 2025.

Thank you!

Questions?